STAFF REVIEW · H446 231 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. No. The returned value is discarded.

2. A variable declared inside a method or block, with restricted scope.

3. Using return.

4. The method can change a caller's variable as a side effect, even if there is no returned result.

I Do — source worked steps

1. Predict first — The function parameter number receives a copy of 6.

2. Model and check — Inside the function number becomes 10; caller x remains 6.

3. Model and check — The parameter leaves scope at the end of the call.

We Do — guided CHECK

1. Model and check
Adding four changes caller y from 9 to 13.

You Do — source mark schemes

1. [3] n is a copy of 7 [1], becomes 9 inside method [1], caller x remains 7 [1].

2. [6] procedure Swap(a:byRef, b:byRef) [1]; temp=a [1]; a=b [1]; b=temp [1]; caller variables must be updated rather than local copies [2].

3. [6] Return version computes a result [1] and caller controls assignment [1]; ref version can update x in place [1]; returning avoids hidden state changes [1], is easier to reason about [1] and can be tested using input/output assertions [1].

Exit Ticket — source answer

1. False. A copy is passed unless a reference-passing mechanism such as ref is used.